

```
|.....|
0x00000020  00 00 00 00 02 03 A5 56 41 6C 1C 67 15 FE F6 9F
|.....VAL.g...|
0x00000030  B1 BD 76 BC 61 BC DE 44 9B 10 45 F3 67 C7 EB 55
|..v.a..D..E.g..U|
0x00000040  9C C3 14 96 E0 A2 39 0C BB 9B CA 87 22 19 A7 87
|.....9....."....|
0x00000050  1E 22 B4 38 16 58 A8 07 AB 58 C2 07 0E 23 27 95
|..".8.X...X...#'.|
0x00000060  0C DA 66 96 CA 15 0B A7 D5 DA 4E 73 70 BC A6 A5
|..f.....Nsp...|
...
```

Display of selective lines

You can filter the hexdump output based on colour code bytes as illustrated below:

Colour	Description
Red	Bytes different in all files
Blue	Bytes different in some files
Green	Bytes same in all files

```
$ binwalk -W --blue dd-wrt.v24_whr-g125.bin
OFFSET      dd-wrt.v24_whr-g125.bin
-----
*
$ binwalk -W --green dd-wrt.v24_whr-g125.bin
OFFSET      dd-wrt.v24_whr-g125.bin
-----
*
$ binwalk -W --red dd-wrt.v24_whr-g125.bin
OFFSET      dd-wrt.v24_whr-g125.bin
-----
0x00000000  48 44 52 30 00 10 36 00 8C 2C 47 85 00 00 01 00
|HDR0..6..G.....|
0x00000010  1C 00 00 00 E4 08 00 00 CC E7 08 00 1F 8B 08 00
|.....|
0x00000020  00 00 00 00 02 03 A5 56 41 6C 1C 67 15 FE F6 9F
|.....VAL.g...|
0x00000030  B1 BD 76 BC 61 BC DE 44 9B 10 45 F3 67 C7 EB 55
|..v.a..D..E.g..U|
0x00000040  9C C3 14 96 E0 A2 39 0C BB 9B CA 87 22 19 A7 87
|.....9....."....|
...
```

Raw LZMA compression stream

The ‘-Z’ option identifies raw LZMA compression data streams from the firmware file as shown below:

```
$ binwalk -Z dd-wrt.v24_whr-g125.bin
```

DECIMAL	HEXADECIMAL	DESCRIPTION
2289	0x8F1	Raw LZMA compression stream, properties: 0x5D [pb: 2, lp: 0, lc: 0], dictionary size: 1048576

Partial compression streams

You can search only for compression streams using the ‘-partial’ option to speed the LZMA scan as follows:

```
$ binwalk --partial -Z dd-wrt.v24_whr-g125.bin
```

DECIMAL	HEXADECIMAL	DESCRIPTION
2289	0x8F1	Raw LZMA compression stream, properties: 0x5D [pb: 2, lp: 0, lc: 0], dictionary size: 33554432
583747	0x8E843	Raw LZMA compression stream, properties: 0x5D [pb: 2, lp: 0, lc: 0], dictionary size: 33554432
607501	0x9450D	Raw LZMA compression stream, properties: 0x5D [pb: 2, lp: 0, lc: 0], dictionary size: 33554432
633288	0x9A9C8	Raw LZMA compression stream, properties: 0x5D [pb: 2, lp: 0, lc: 0], dictionary size: 33554432
658249	0xA0B49	Raw LZMA compression stream, properties: 0x5D [pb: 2, lp: 0, lc: 0], dictionary size: 33554432
...		

Length limit

The ‘-length’ option is used to limit the size of bytes in the firmware file for analysis. For example:

```
$ binwalk --length=0x500 dd-wrt.v24_whr-g125.bin
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	TRX firmware header, little endian, image size: 3543040 bytes, CRC32: 0x85472C8C, flags: 0x0, version: 1, header size: 28 bytes, loader offset: 0x1C, linux kernel offset: 0x8E4, rootfs offset: 0x8E7CC
28	0x1C	gzip compressed data, maximum compression, from Unix, last modified: 1970-01-01 00:00:00 (null date)

Opcodes

The CPU architecture opcodes can be viewed using the ‘-A’ option as indicated below:

```
$ binwalk -A dd-wrt.v24_whr-g125.bin
```